Improved Hip Fracture Prediction Using 3D-DXA–Based Finite Element Models in the JPOS Cohort

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Introduction

Areal bone mineral density (aBMD) derived from dual-energy X-ray absorptiometry (DXA) is the clinical gold standard for diagnosing osteoporosis. Quantitative computed tomography (QCT)-based finite element (FE) simulations can improve hip fracture prediction compared with aBMD, but need for QCT scans limits clinical adoption. Advances in 2D–3D modelling enable the generation of patient-specific 3D-DXA models from standard DXA images, offering potential for fracture risk prediction when combined with 3D FE analyses of osteoporosis patient cohorts [1]. This study aimed to evaluate whether the exploitation of 3D-DXA-FE simulations for fracture prediction can be extended to population cohorts.

Methods

We conducted a case–control study including 1,636 women of the JPOS Study, 85 of whom suffered a hip fracture over 20-year follow-up. 3D-DXA-FE models were generated from hip DXA scans out of 3D-Shaper® (v2.14) appearance and shape modelling [2]. Side-fall was simulated by applying displacement to the femoral head under rigid-body constraints, with the greater trochanter and shaft fixed. A perfect elastoplastic material model used bone density-dependent material properties and tension- and compression-specific yield parameters, distributed over the mesh. Failure was defined from the nonlinear force–displacement curve at the onset of force decline. Global mechanical descriptors, including strength, nonlinear deformation, and energy absorption, were quantified from FE analyses and used for fracture discrimination [1]. Figure 1 illustrates the overall methodology.

Results

The area under the receiver operating characteristic curve (AUROC) with leave-one-out cross-validation showed that combining strength and residual energy (AUROC: 0.797) improved fracture prediction compared with total hip aBMD (AUROC: 0.763) (Fig.2). The difference between the two AUROCs was statistically significant according to the DeLong test ($p = 0.0267$).

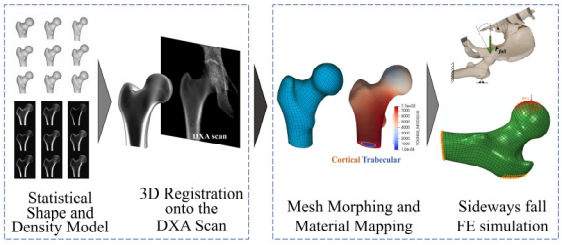


Fig.1. Schematic overview of the conceptual framework used in this study

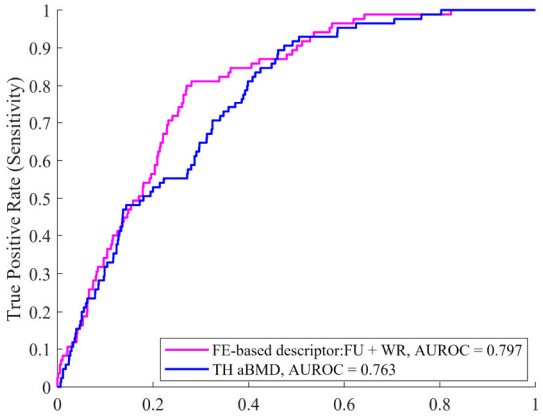


Fig. 2. ROC curves containing the FE-based descriptor.

Conclusions

This is the first study demonstrating that global mechanical descriptors obtained from standard hip DXA scans outperform aBMD in predicting incident hip fractures in a population-based, prospectively followed cohort of women. These findings highlight the potential of integrating 3D-DXA and FE modelling into existing DXA-based clinical workflows to enhance fracture risk assessment.

Acknowledgements

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References

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